

Advanced Python for Data Science

Week(7): Integrated System Design

Friday, 24th July 2026, 9:30-12:45

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What We'll Cover Today

1

Scalable data pipelines

Batch vs. streaming processing

2

Microservices architecture

Independent services over HTTP

3

Performance trade-offs

Sequential vs. concurrent requests

4

Monitoring & logging

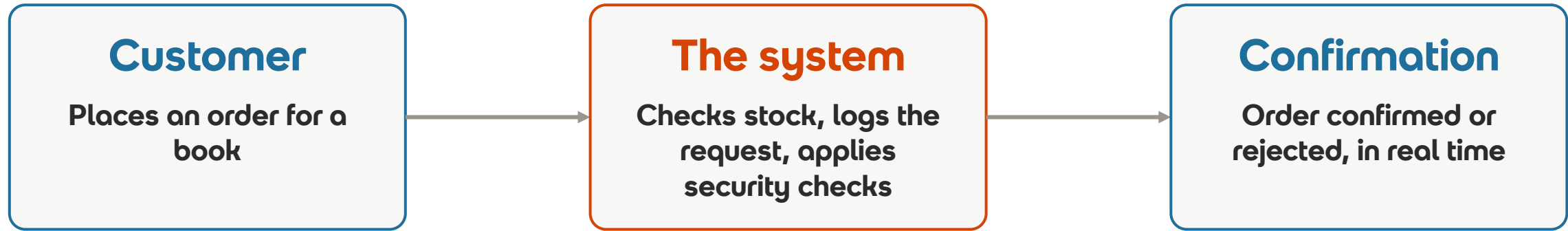
Structured logs and a metrics endpoint

5

Security considerations

API keys and input validation

The example: an online bookstore



Two ways to move orders through the system

Batch



Wait for a group, then process it all at once. Simple, but nothing moves until the group is ready.

Streaming

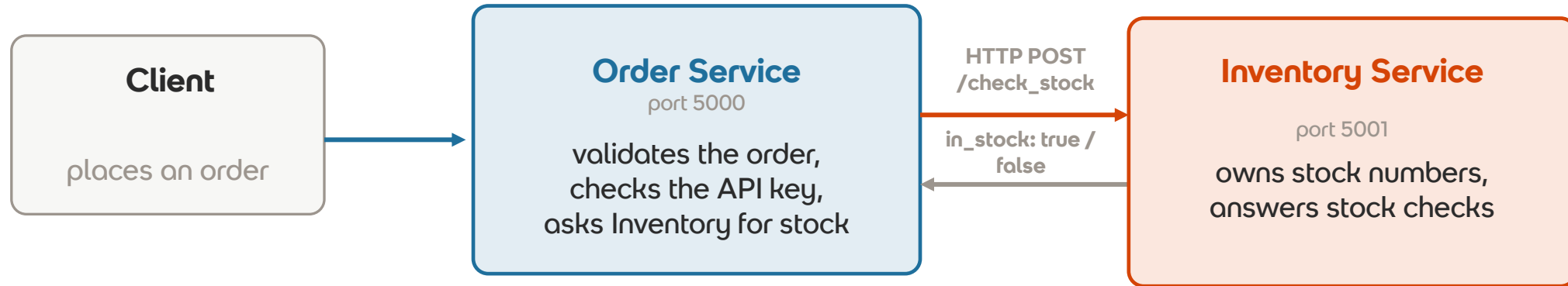


Handle each order the moment it arrives. Reacts instantly, at the cost of a little extra bookkeeping per item.

Question: Which Pipeline works the best for the book online shop?

2. MICROSERVICES ARCHITECTURE

Two small, independent services instead of one large program 



Each service can be built, tested, deployed, and scaled on its own. They only know about each other through their HTTP interface.

Both services also run as Docker containers, wired together with docker-compose.yml → same architecture, one build command.

Sequential vs Concurrent Order Requests

Sequential



One request waits for the previous one to finish before it starts.

Concurrent

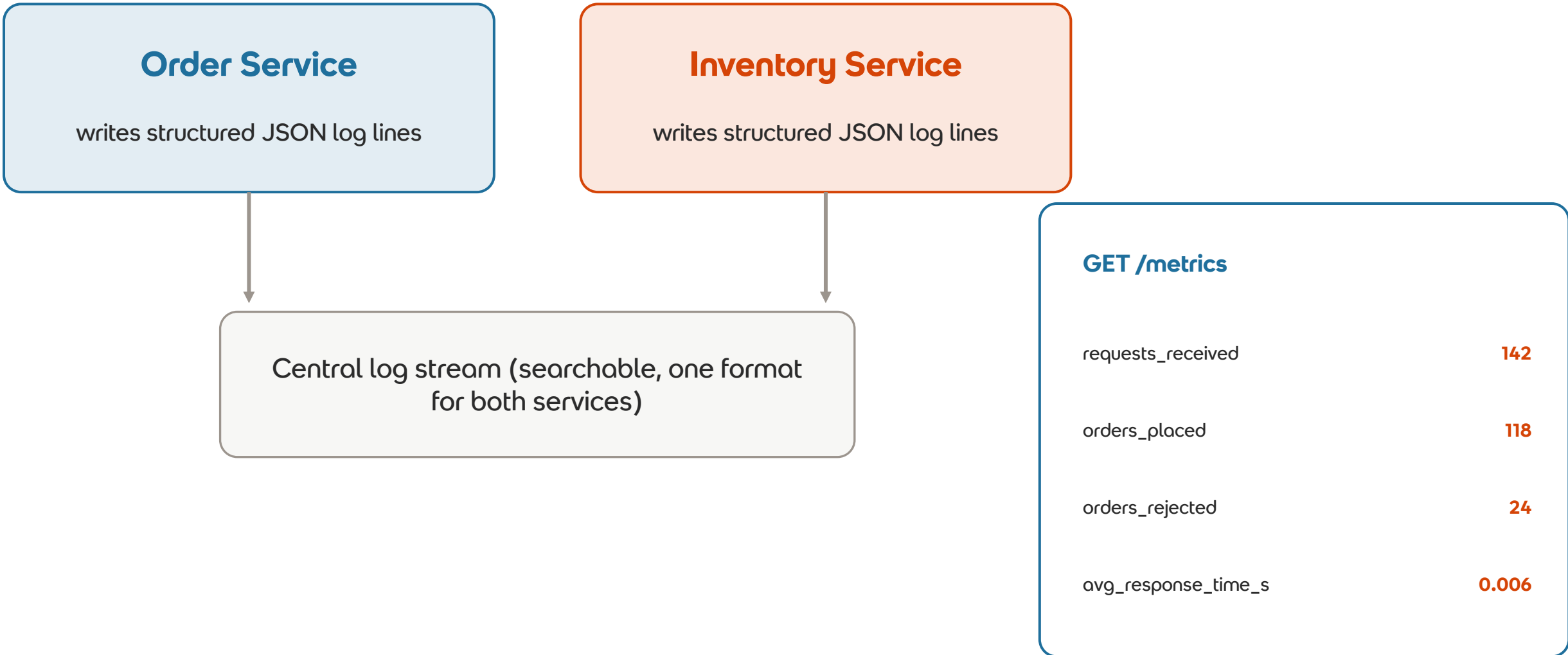


Several requests are in flight at once, overlapping their waiting time.

4. MONITORING & LOGGING

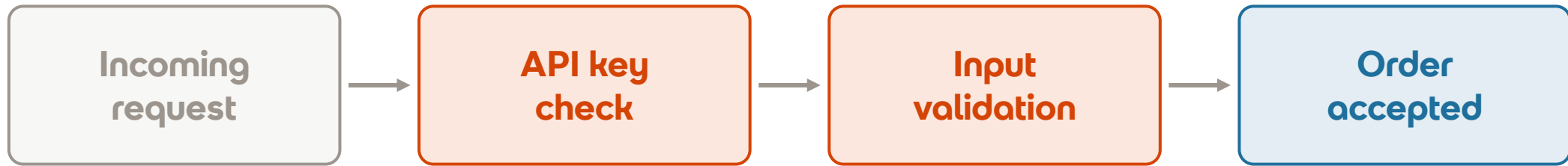


Every service reports what it's doing

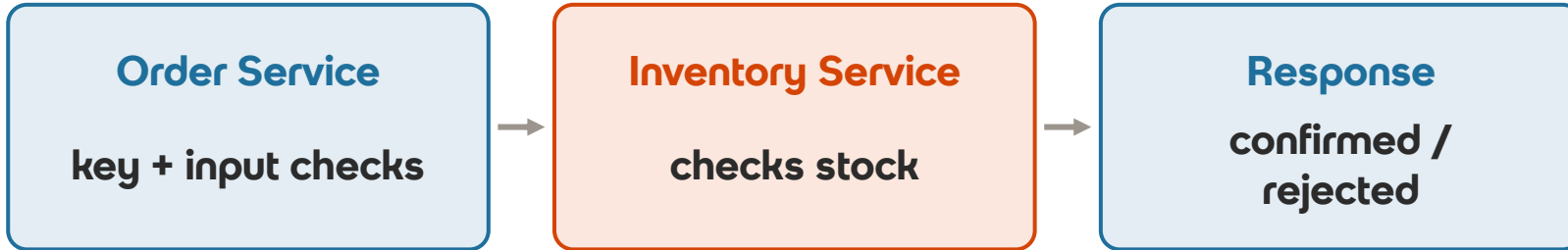


5. SECURITY CONSIDERATIONS

Every incoming order passes through two checks before it is trusted



One order, moving through every layer we covered today



logged at every step · every step measured for /metrics · every step guarded by validation

Concept	In this system
Data pipelines	Orders processed as a stream, not a batch
Microservices	Order Service and Inventory Service, split by responsibility
Performance	Concurrent requests instead of one at a time
Monitoring & logging	Structured logs plus a /metrics endpoint
Security	API key check and input validation at the boundary

EXERCISES



Week7_Exercise.ipynb → Three-Four exercises that add real capability, plus one summer-break activity

1 Add a Notification Service

A third microservice that confirms each order without blocking it

2 Trace a request across services

The same request ID should appear in every service's logs

3 Add a rate limiter

No more than 5 requests per client in a 10-second window

4 Decouple with a queue (Take-home Assignment)

Use Python's queue module so services don't wait on each other

5 Deploy with Kubernetes • Fun Summer-break Activity

Apply your Week 6 Kubernetes skills to the Dockerized services from this week → write Deployments and Services for both.

Check the starter code in `Week7.ipynb`

You have 45 minutes to do so

Pause / Break

See you in 15 minutes 😊!
Please be on time.

Exercises

Solve Problems (1 -3) in "Week7_Exercise.ipynb". *Don't forget to split your screen. Read the whole instructions first and answer the reflection questions in your own words.*

Week 7 Extension Exercises: Integrated System Design

- These exercises build directly on the bookstore system from the starter notebook (`order_service.py` and `inventory_service.py`).
- Work through them in order. Each one adds a real capability that a production system would need.

Before starting, make sure (`order_service.py` and `inventory_service.py`) are in the same folder as this notebook.

```
1 import subprocess
2 import time
3 import requests
4
5 inventory_process = subprocess.Popen(["python3", "inventory_service.py"])
6 time.sleep(1.5)
7 order_process = subprocess.Popen(["python3", "order_service.py"])
8 time.sleep(1.5)
9
10 API_KEY = "week7-secret-key"
11 print(requests.get("http://localhost:5001/health").json())
12 print(requests.get("http://localhost:5000/health").json())
```

```
{'service': 'inventory_service', 'status': 'ok'}
{'service': 'order_service', 'status': 'ok'}
```

Problem 1: Add a Notification Service

Any Questions?

See you next time 😊

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